# Supplementary Online Content

Sati N, Boyne D, Cheung WY, Cash SB, Arora P. Factors modifying the associations of single or combination programmed cell death 1 and programmed cell death ligand 1 inhibitor therapies with survival outcomes in patients with metastatic clear cell renal cell carcinoma: a systematic review and meta-analysis. *JAMA Netw Open*. 2021;4(1):e2034201. doi:10.1001/jamanetworkopen.2020.34201

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**This supplementary material has been provided by the authors to give readers additional information about their work.**

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**eTable 1.** MEDLINE search strategy

| MEDLINE Search Strategy | | | |
|---|---|---|---|
| **Concept** | **Line** | **Search Term** | **Hits** |
| **Cancer** | 1 | Carcinoma, Renal Cell/ | 31607 |
| | 2 | ((kidney* or renal or hypernephroid or nephroid) adj2 (tumo?r* or cancer* or neoplas* or carcinoma* or adenocarcinoma*)).ti,ab,kw,kf. | 62966 |
| | 3 | hypernephroma*.ti,ab,kw,kf. (1368) | 1368 |
| | 4 | 1 or 2 or 3 (66886) | **66998** |
| **Intervention** | 5 | antibodies, monoclonal, humanized/ (34179) | 34235 |
| | 6 | human* monoclonal antibod*.ti,ab,kw,kf. (6735) | 6749 |
| | 7 | programmed cell death 1 receptor/ (4998) | 5047 |
| | 8 | programmed cell death 1 receptor*.ti,ab,kw,kf. (101) | 101 |
| | 9 | B7-H1 antigen/ (4339) | 4381 |
| | 10 | immunotherapy/ (41792) | 41887 |
| | 11 | checkpoint inhibit*.ti,ab,kw,kf. (7469) | 7573 |
| | 12 | Pembrolizumab.ti,ab,kw,kf. (2493) | 2525 |
| | 13 | nivolumab/ (1654) | 1686 |
| | 14 | nivolumab.ti,ab,kw,kf. (3438) | 3486 |
| | 15 | avelumab.ti,ab,kw,kf. (270) | 275 |
| | 16 | atezolizumab.ti,ab,kw,kf. (608) | 613 |
| | 17 | durvalumab.ti,ab,kw,kf. (290) | 291 |
| | 18 | 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 (92600) | **92880** |
| **Study Design** | 19 | clinical trial/ or clinical trial, phase ii/ or clinical trial, phase iii/ or randomized controlled trial/ (804532) | 805134 |
| | 20 | randomized controlled trial.pt. (487883) | 488336 |
| | 21 | random*.mp. (1295626) | 1297771 |
| | 22 | random allocation/ (100129) | 100224 |
| | 23 | double-blind method/ or single-blind method/ (179174) | 179347 |
| | 24 | ((singl* or doubl* or tripl*) adj3 (blind* or mask*)).mp. (228979) | 229236 |
| | 25 | 19 or 20 or 21 or 22 or 23 or 24 (1615878) | 1618194 |
| | 26 | 4 and 18 and 25 (447) | **450** |
| **Final** | 27 | limit 26 to (english language and yr="2006 -Current | **273** |

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**eTable 2.** Cochrane search strategy

| Cochrane Search Strategy | | | |
|---|---|---|---|
| **Concept** | **Line** | **Search Term** | **Hits** |
| **Cancer** | 1 | MeSH descriptor: [Carcinoma, Renal Cell] explode all trees | 825 |
| | 2 | (((kidney* or renal or hypernephroid or nephroid) near/2 (tumo*r* or cancer* or neoplas* or carcinoma* or adenocarcinoma*))):ti,ab,kw | 3746 |
| | 3 | (hypernephroma*):ti,ab,kw | 6 |
| | 4 | #1 or #2 or #3 | **3749** |
| **Intervention** | 5 | MeSH descriptor: [Antibodies, Monoclonal, Humanized] explode all trees | 6231 |
| | 6 | (human* monoclonal antibod*):ti,ab,kw | 11676 |
| | 7 | MeSH descriptor: [Programmed Cell Death 1 Receptor] explode all trees | 48 |
| | 8 | programmed cell death 1 receptor*:ti,ab,kw | 499 |
| | 9 | MeSH descriptor: [B7-H1 Antigen] explode all trees | 39 |
| | 10 | MeSH descriptor: [Immunotherapy] explode all trees | 7691 |
| | 11 | (checkpoint inhibit*):ti,ab,kw | 1028 |
| | 12 | (pembrolizumab):ti,ab,kw | 1069 |
| | 13 | MeSH descriptor: [Nivolumab] explode all trees | 135 |
| | 14 | (nivolumab):ti,ab,kw | 1334 |
| | 15 | (avelumab):ti,ab,kw | 149 |
| | 16 | (atezolizumab):ti,ab,kw | 502 |
| | 17 | (durvalumab):ti,ab,kw | 345 |
| | 18 | #5 or #6 or #7 or #8 or #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 | **24596** |
| **Study Design** | 19 | MeSH descriptor: [Clinical Trial] explode all trees | 147 |
| | 20 | ("randomized controlled trial"):pt | 476595 |
| | 21 | (random*) | 1072167 |
| | 22 | MeSH descriptor: [Random Allocation] explode all trees | 20592 |
| | 23 | MeSH descriptor: [Double-Blind Method] explode all trees | 132487 |
| | 24 | MeSH descriptor: [Single-Blind Method] explode all trees | 19558 |
| | 25 | ((singl* or doubl* or tripl*) near/3 (blind* or mask*)) | 347817 |
| | 26 | #19 or #20 or #21 or #22 or #23 or #24 or #25 | **1112537** |
| **Final** | 27 | #4 and #18 and #26 with Publication Year from 2006 to 2019, in Trials | **389** |

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**eTable 3.** Inclusion/Exclusion Criteria

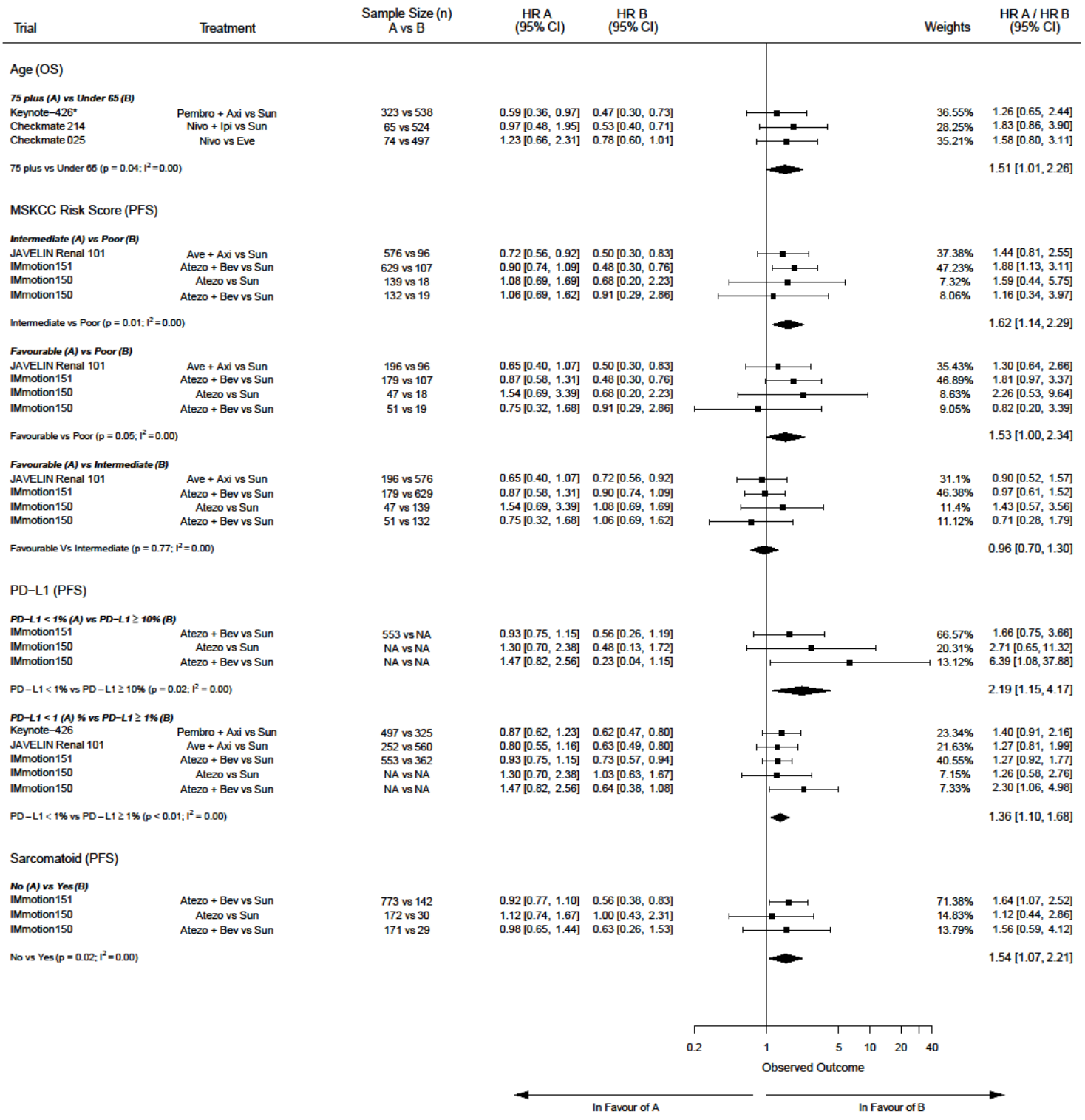
| | INCLUDE | EXCLUDE |
|---|---|---|
| **POPULATION** | • Human patients with stage IV renal cell carcinoma | • Patients with stage I-III renal cell carcinoma<br>• Laboratory studies done in cell lines or animals |
| **INTERVENTION** | • Anti-PD-1/PD-L1 monotherapies<br>  ○ Pembrolizumab (anti-PD-1)<br>  ○ Nivolumab (anti-PD-1)<br>  ○ Avelumab (anti-PD-L1)<br>  ○ Atezolizumab (anti-PD-L1)<br>  ○ Durvalumab (anti-PD-L1)<br>  ○ Other anti-PD-1/PD-L1 inhibitors<br>*- OR -*<br>• Any of the above anti-PD-1/PD-L1 therapies **in combination** with other immunotherapies | |
| **COMPARATOR** | • Any patient, tumor, or treatment characteristics<br>• Types of subgroups:<br>  ○ Age<br>  ○ Sex<br>  ○ Ethnicity<br>  ○ Genetics<br>  ○ Geographic location<br>  ○ Disease severity<br>  ○ Health-related quality of life (HRQoL)<br>  ○ Other subgroups | • Studies that do not report a subgroup analysis or a regression analysis that includes an interaction term between the treatment and any other variable of interest |
| **OUTCOME** | • Any survival outcomes (e.g. overall, progression-free, disease-free, metastasis-free survival) | • Studies that do not report on a survival outcome |
| **TIMING** | • Predictive factors measured at or prior to initiation of the immunotherapies<br>• Survival outcomes measured from the date of randomization, diagnosis, or chemotherapy initiation<br>• Studies conducted since 2006 | • Studies conducted prior to 2006 |
| **STUDY DESIGN** | • Phase 2 or phase 3 randomized controlled trials | • Phase 1 or phase 4 RCTs<br>• Non-interventional studies<br>• Non-randomized studies<br>  ○ Observational studies<br>  ○ Reviews<br>  ○ Case-studies<br>• Non-English studies |

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**eTable 4.** PD-L1 expression and median survival outcomes across trials. Data that have not been reached or are non-estimable have been identified as NR or NE, respectively

| Trial | Intervention | Comparator | PD-L1 | Intervention Median OS, in Months (95% CI) | Comparator Median OS, in Months (95% CI) | Intervention Median PFS, in Months (95% CI) | Comparator Median PFS, in Months (95% CI) |
|---|---|---|---|---|---|---|---|
| NCT01354431[34] | Nivolumab | N/A | <5% | 18.2 (12.7-26) | NA | 2.9 (2.1-4.2) | - |
| | Nivolumab | N/A | ≥5% | NR (13.4-NR) | NA | 4.9 (1.4-7.8) | - |
| IMmotion150[35] | Atezo + Bev | Sun | <1% | - | - | 11.7 (8.4-17.3) | 8.4 (7-14) |
| | Atezo | Sun | <1% | - | - | 6.1 (5.4-13.6) | 8.4 (7-14) |
| | Atezo + Bev | Sun | ≥1% | - | - | 14.7 (8.2-25.1) | 7.8 (3.8-10.8) |
| | Atezo | Sun | ≥1% | - | - | 5.5 (3-13.9) | 7.8 (3.8-10.8) |
| Checkmate 025[36] | Nivo | Eve | <1% | 27.4 (21.4-NE) | 21.2 (17.7-26.2) | - | - |
| | Nivo | Eve | ≥1% | 21.8 (16.5-28.1) | 18.8 (11.9-19.9) | - | - |
| | Nivo | Eve | <5% | 24.6 (21.4-NE) | 20 (17.7-24.7) | - | - |
| | Nivo | Eve | ≥5% | 21.9 (14-NE) | 18.1 (10.3-NE) | - | - |
| IMmotion151[38] | Atzeo + Bev | Sun | <1% | NE (NE-NE) | NE (23.3-NE) | 11.2 (9.6-13.6) | 8.4 (7.5-9.7) |
| | Atezo + Bev | Sun | ≥1% | NE (NE-NE) | 23.3 (21.3-NE) | 11.2 (8.6-14.3) | 7.7 (6.8-9.7) |

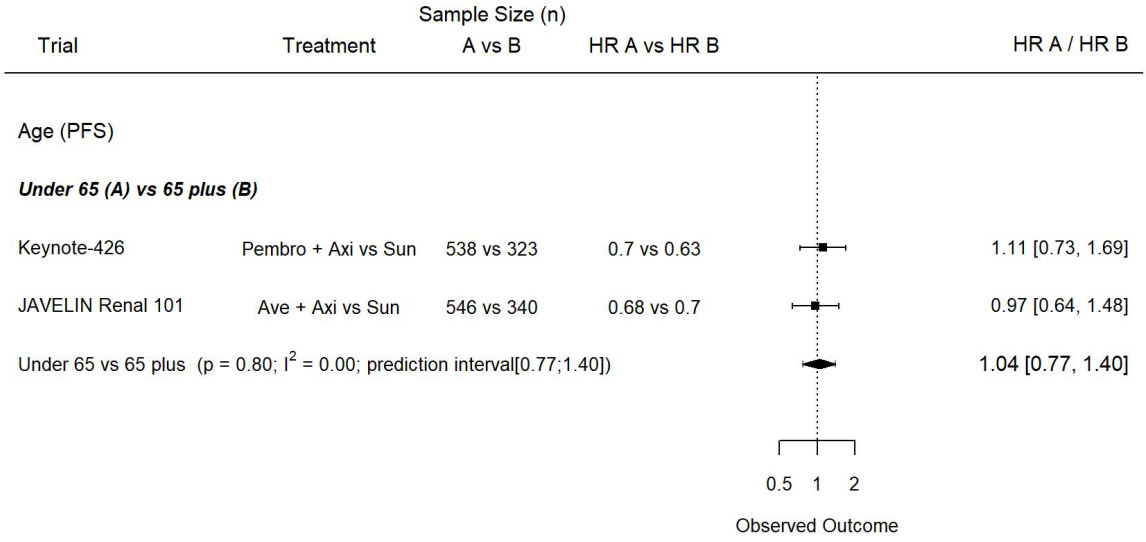
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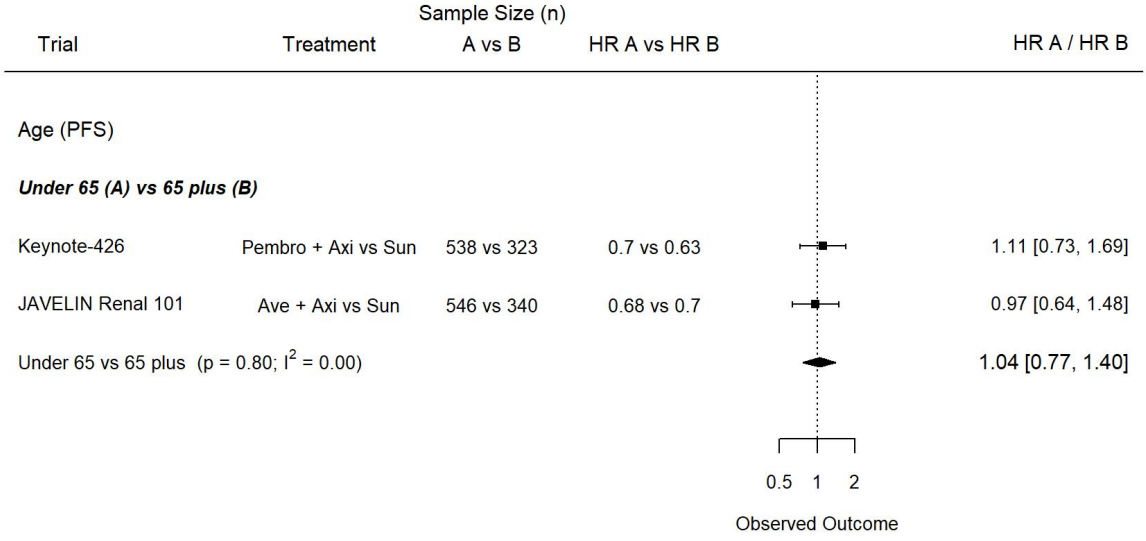

**eFigure 1.** Fixed−effects meta−analyses on the ratio of subgroup−specific hazard ratios among metastatic clear−cell renal cell carcinoma patients treated with anti−PD−1/PD−L1 therapies.
*Older age subgroup of ≥ 65
**Not statistically significant

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## Age (PFS)



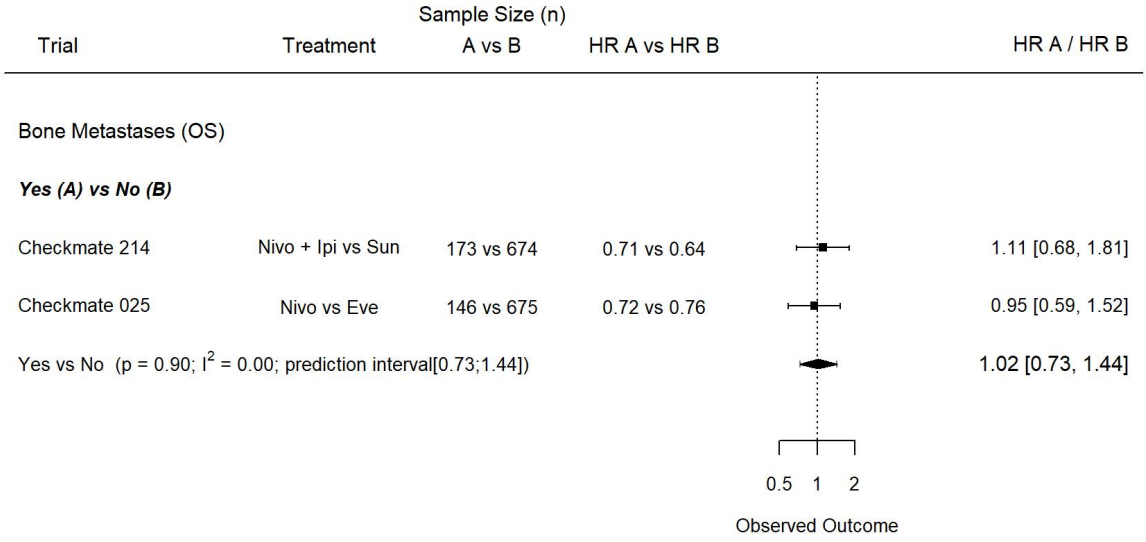

eFigure 2. Random-effects meta-analysis on the ratio of subgroup-specific hazard ratios for age ($HR_{age \geq 65}$ to $HR_{age < 65}$=1.04; 95% CI:0.77-1.40; 95% PI: 0.77-1.40; $I^2$=0%; *p*=0.8).



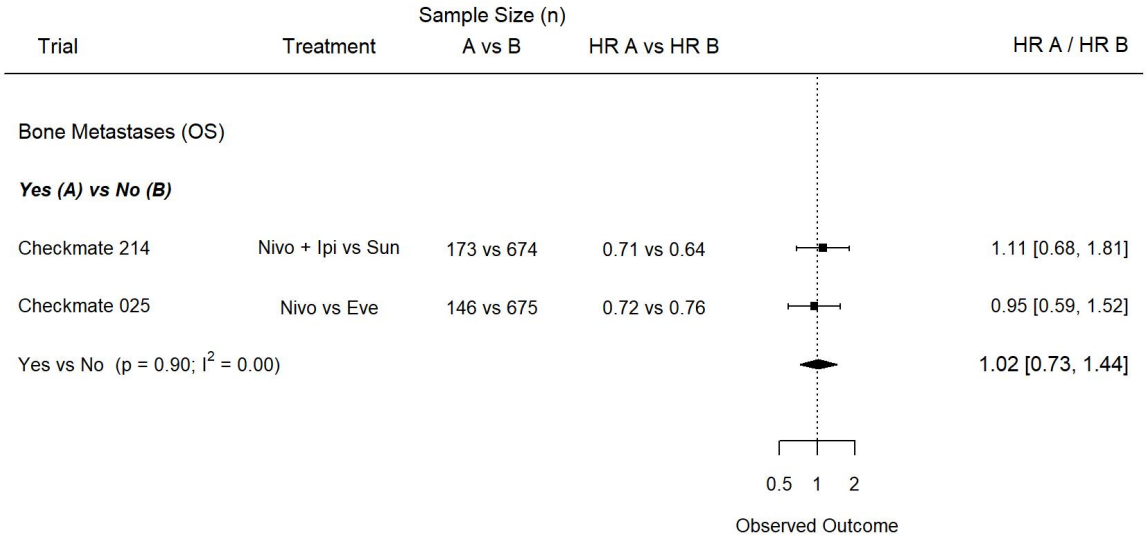

eFigure 3. Fixed-effects meta-analysis on the ratio of subgroup-specific hazard ratios for age ($HR_{age \geq 65}$ to $HR_{age < 65}$=1.04; 95% CI:0.77-1.40; $I^2$=0%; *p*=0.8).

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## Bone Metastases (OS)



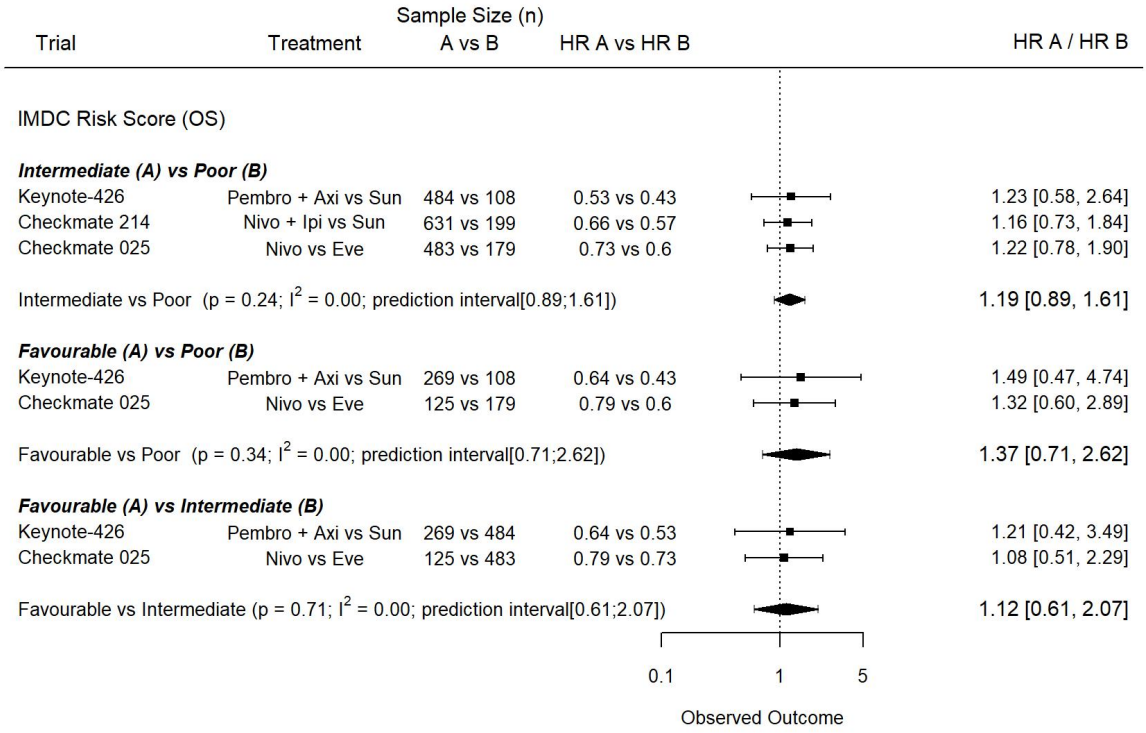

eFigure 4. Random-effects meta-analysis on the ratio of subgroup-specific hazard ratios for bone mestastases ($HR_{Yes}$ to $HR_{No}$=1.02; 95% CI:0.73-1.44; 95% PI: 0.73-1.44; $I^2$=0%; *p*=0.9).




eFigure 5. Fixed-effects meta-analysis on the ratio of subgroup-specific hazard ratios for bone mestastases ($HR_{Yes}$ to $HR_{No}$=1.02; 95% CI:0.73-1.44; $I^2$=0%; *p*=0.9).

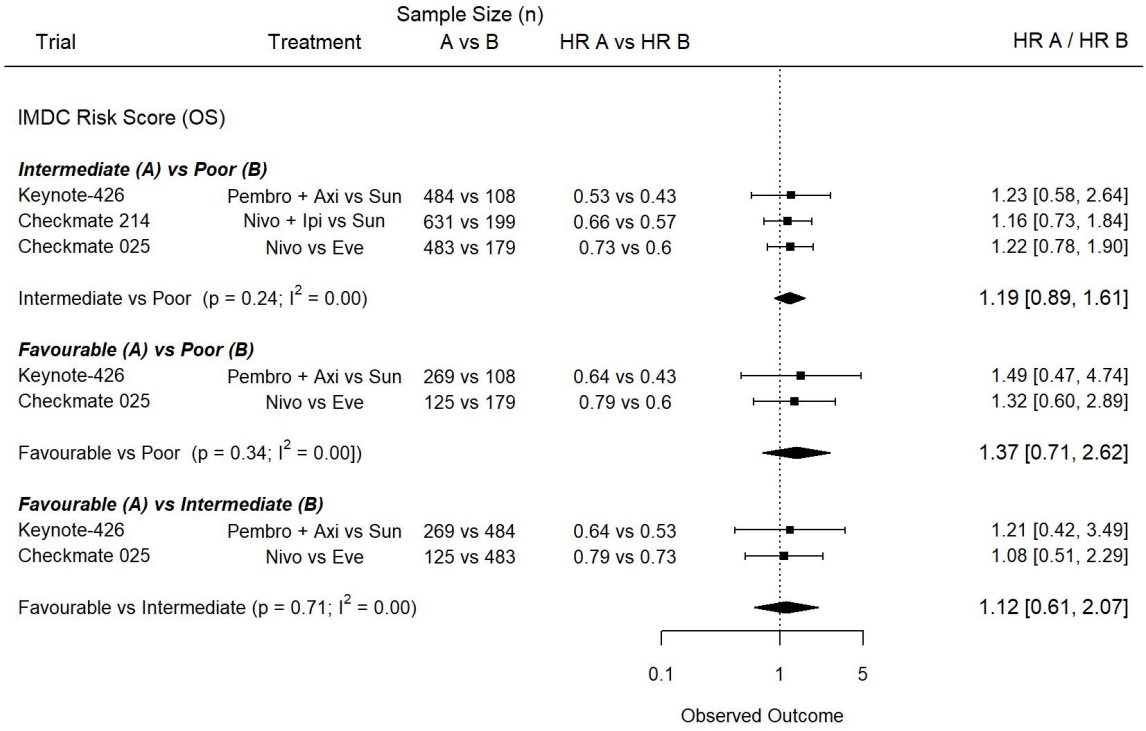
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## IMDC Risk Score (OS)




eFigure 6. Random-effects meta-analysis on the ratio of subgroup-specific, overall-survival hazard ratios for IMDC risk score ($HR_{Intermediate}$ to $HR_{Poor}$=1.19; 95% CI:0.89-1.61; 95% PI:0.89-1.61; $I^2$=0%; *p*=0.24; $HR_{Favourable}$ to $HR_{Poor}$=1.37; 95% CI:0.71-2.62; 95% PI:0.71-2.62; $I^2$=0%; *p*=0.34; $HR_{Favourable}$ to $HR_{Intermediate}$=1.12; 95% CI:0.61-2.07; 95% PI:0.61-2.07; $I^2$=0%; *p*=0.71).

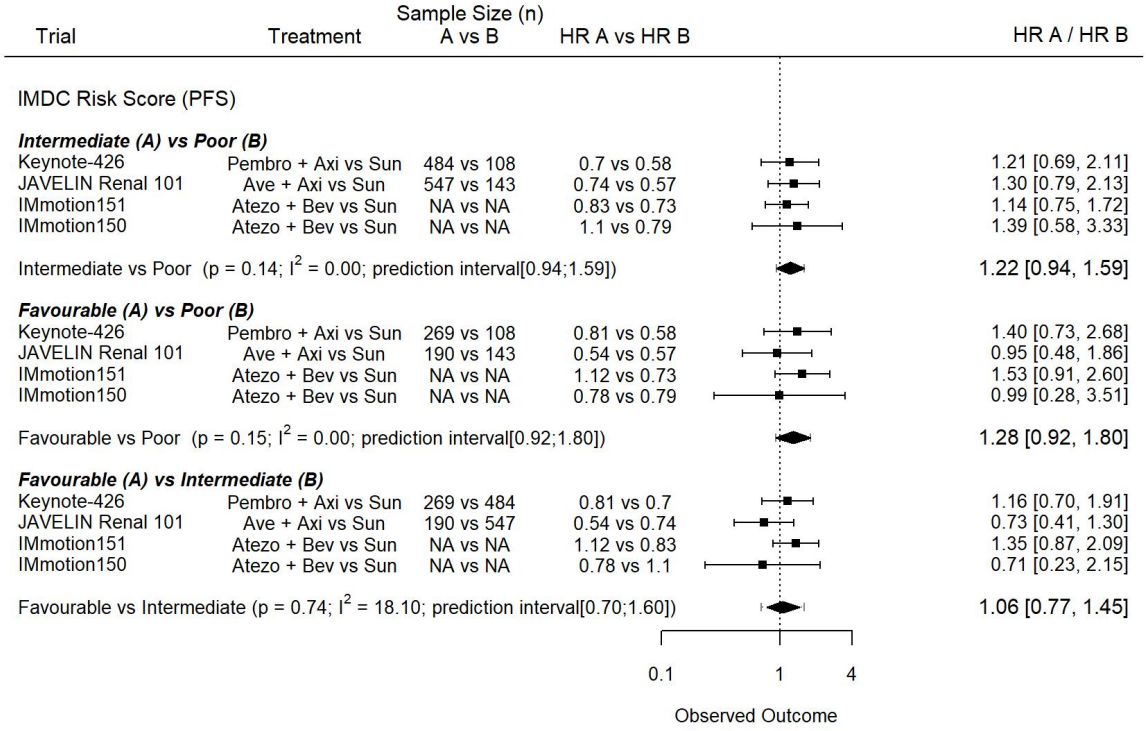
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eFigure 7. Fixed-effects meta-analysis on the ratio of subgroup-specific, overall-survival hazard ratios for IMDC Risk Score ($HR_{Intermediate}$ to $HR_{Poor}$=1.19; 95% CI:0.89-1.61; $I^2$=0%; *p*=0.24; $HR_{Favourable}$ to $HR_{Poor}$=1.37; 95% CI:0.71-2.62; $I^2$=0%; *p*=0.34; $HR_{Favourable}$ to $HR_{Intermediate}$=1.12; 95% CI:0.61-2.07; $I^2$=0%; *p*=0.71).

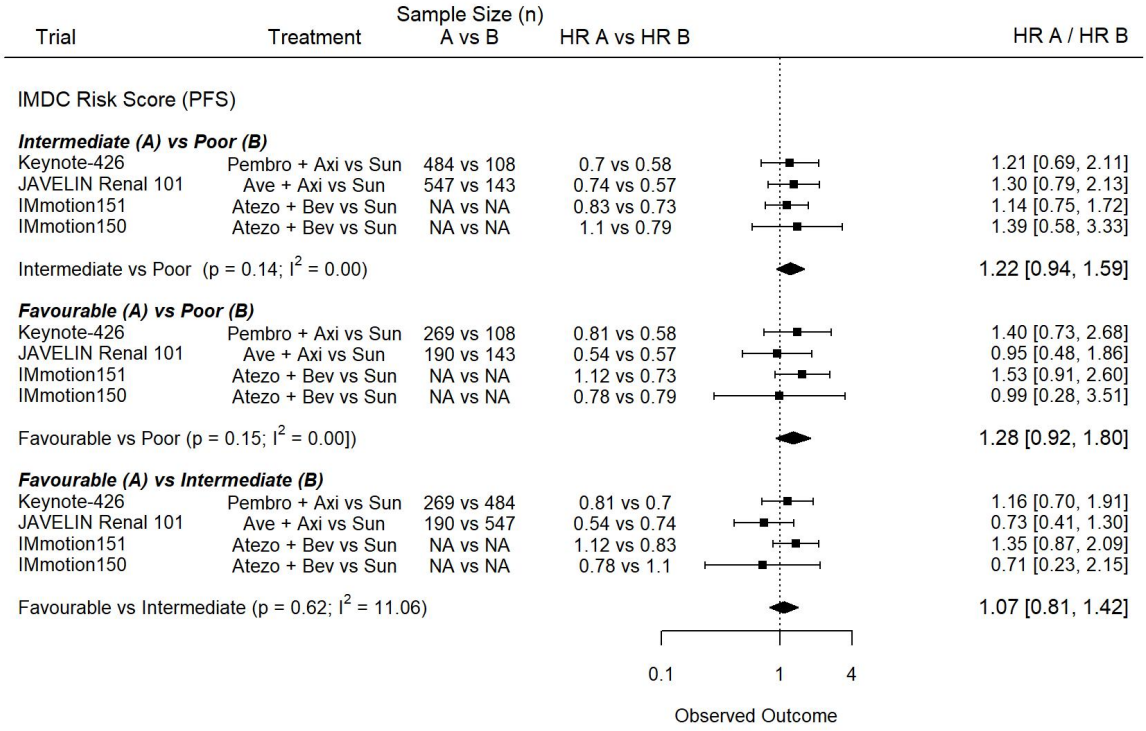
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## IMDC Risk Score (PFS)




eFigure 8. Random-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for IMDC risk score ($HR_{Intermediate}$ to $HR_{Poor}$=1.22; 95% CI:0.94-1.59; 95% PI:0.94-1.59; $I^2$=0%; *p*=0.14; $HR_{Favourable}$ to $HR_{Poor}$=1.28; 95% CI:0.92-1.80; 95% PI:0.92-1.80; $I^2$=0%; *p*=0.15; $HR_{Favourable}$ to $HR_{Intermediate}$=1.06; 95% CI:0.77-1.45; 95% PI:0.70-1.60; $I^2$=18.10%; *p*=0.74).

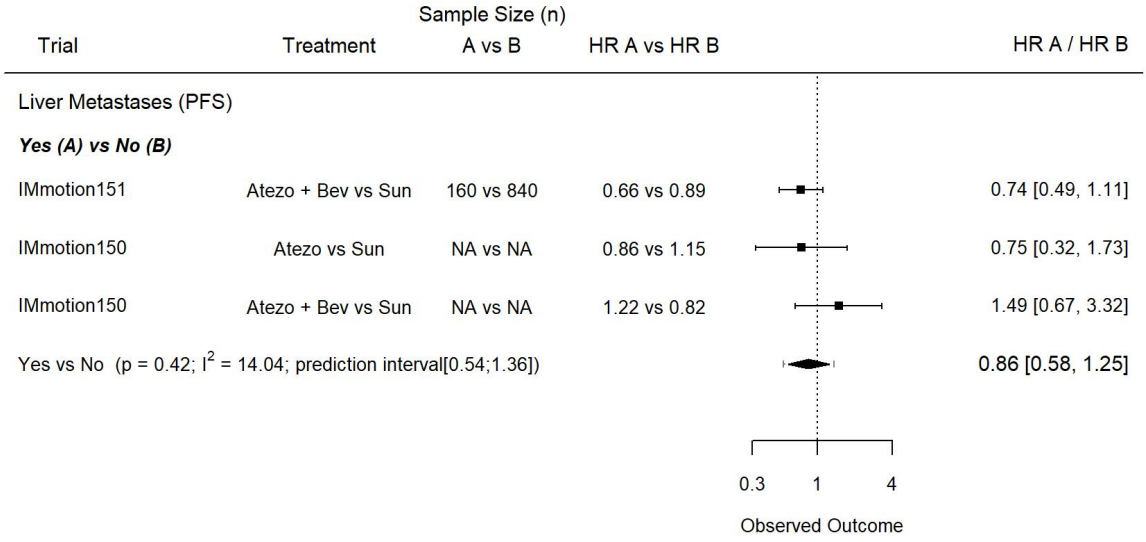
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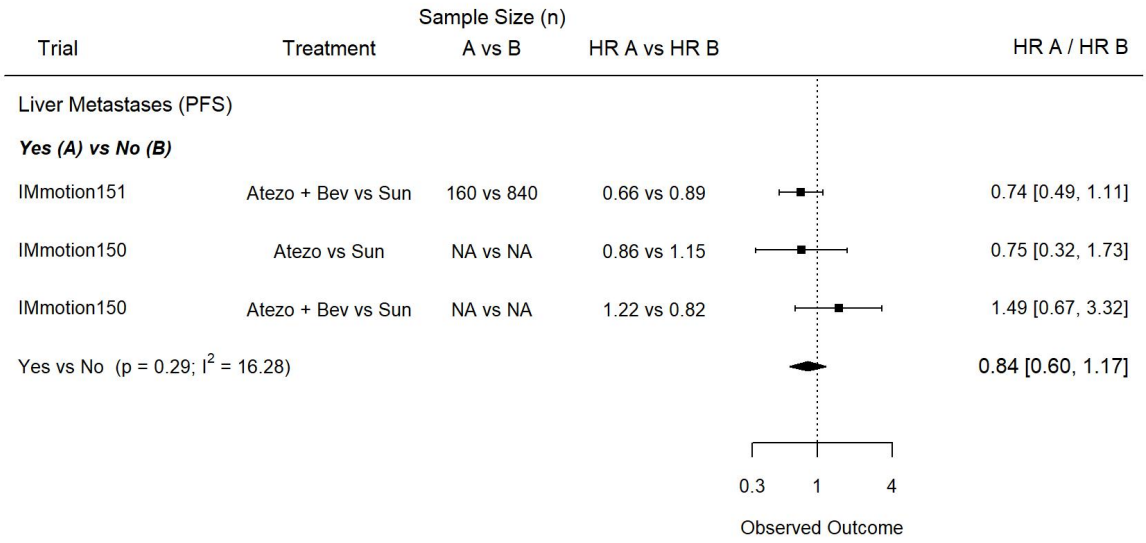

eFigure 9. Fixed-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for IMDC risk score ($HR_{Intermediate}$ to $HR_{Poor}$=1.22; 95% CI:0.94-1.59; $I^2$=0%; *p*=0.14; $HR_{Favourable}$ to $HR_{Poor}$=1.28; 95% CI:0.92-1.80; $I^2$=0%; *p*=0.15; $HR_{Favourable}$ to $HR_{Intermediate}$=1.07; 95% CI:0.81-1.42; $I^2$=11.06%; *p*=0.62).

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## Liver Metastases (PFS)



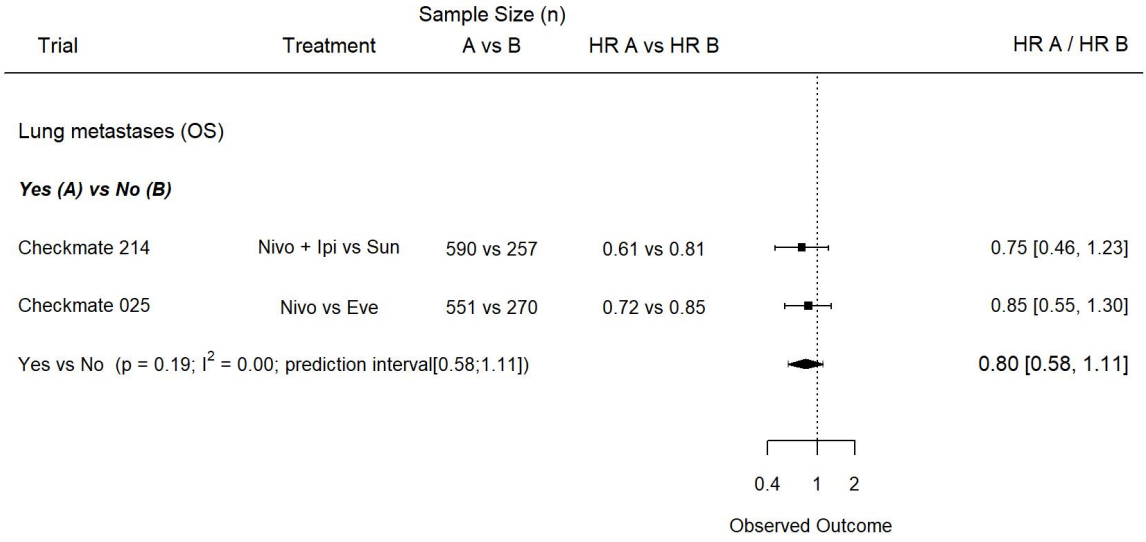

eFigure 10. Random-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for liver metastases ($HR_{Yes}$ to $HR_{No}$=0.86; 95% CI:0.58-1.25; 95% PI:0.54-1.36; $I^2$=14.04%; *p*=0.42).



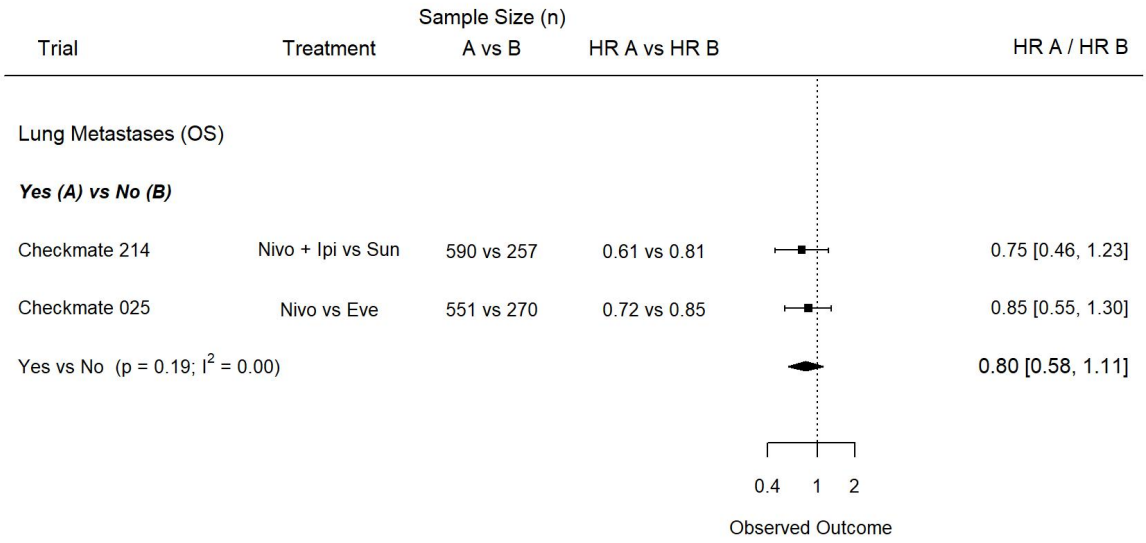

eFigure 11. Fixed-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for liver metastases ($HR_{Yes}$ to $HR_{No}$=0.84; 95% CI:0.60-1.17; $I^2$=16.28%; *p*=0.29).

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## Lung Metastases (OS)



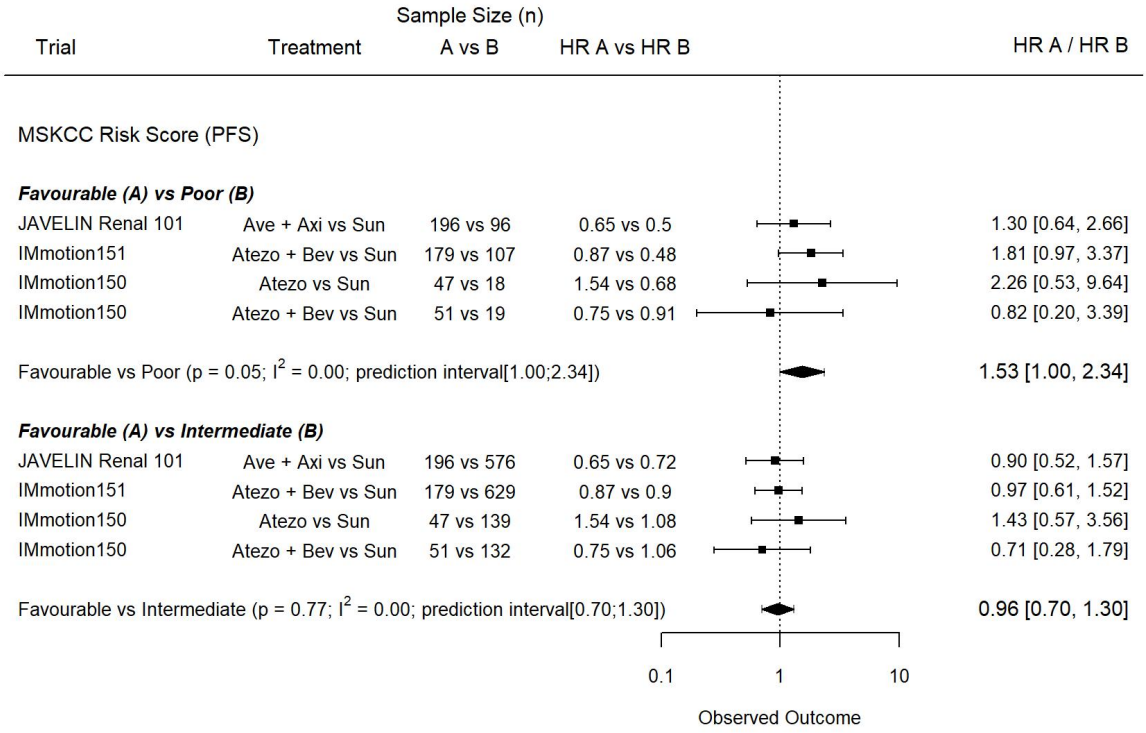

eFigure 12. Random-effects meta-analysis on the ratio of subgroup-specific, overall-survival hazard ratios for lung metastases ($HR_{Yes}$ to $HR_{No}$=0.80; 95% CI:0.58-1.11; 95% PI:0.58-1.11; $I^2$=0%; *p*=0.19).




eFigure 13. Fixed-effects meta-analysis on the ratio of subgroup-specific, overall-survival hazard ratios for lung metastases ($HR_{Yes}$ to $HR_{No}$=0.80; 95% CI:0.58-1.11; $I^2$=0%; *p*=0.19).

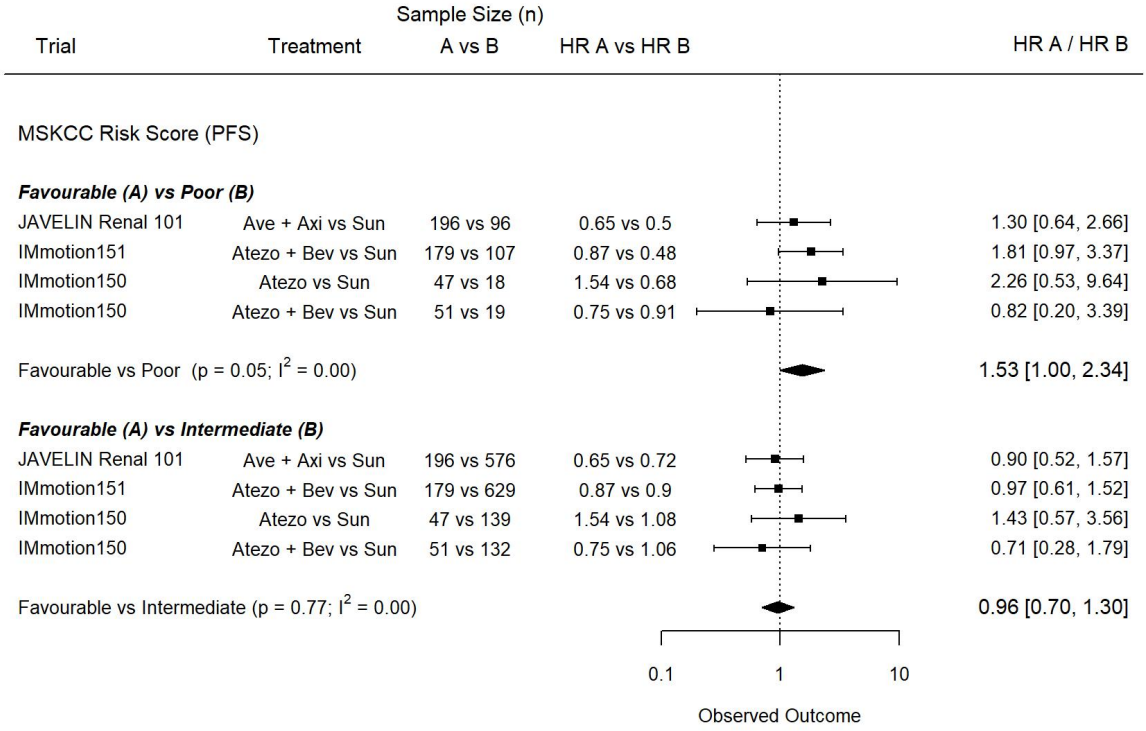
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## MSKCC Risk Score (PFS)




eFigure 14. Random-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for MSKCC Risk Score ($HR_{Favourable}$ to $HR_{Poor}$=1.53; 95% CI:1.00-2.34; 95% PI: 1.00-2.34; $I^2$=0%; *p*=0.05. $HR_{Favourable}$ to $HR_{Intermediate}$=0.96; 95% CI:0.70-1.30; 95% PI:0.70-1.30; $I^2$=0%; *p*=0.77).

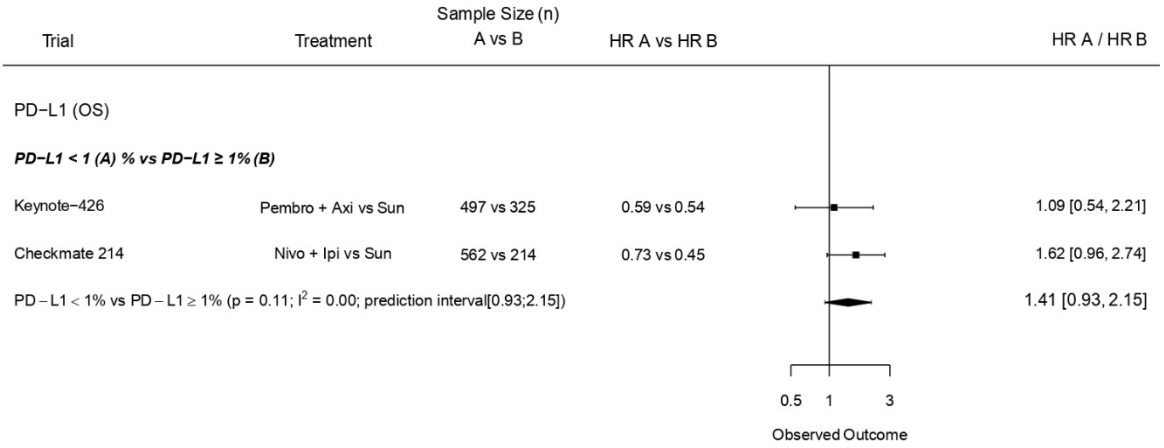
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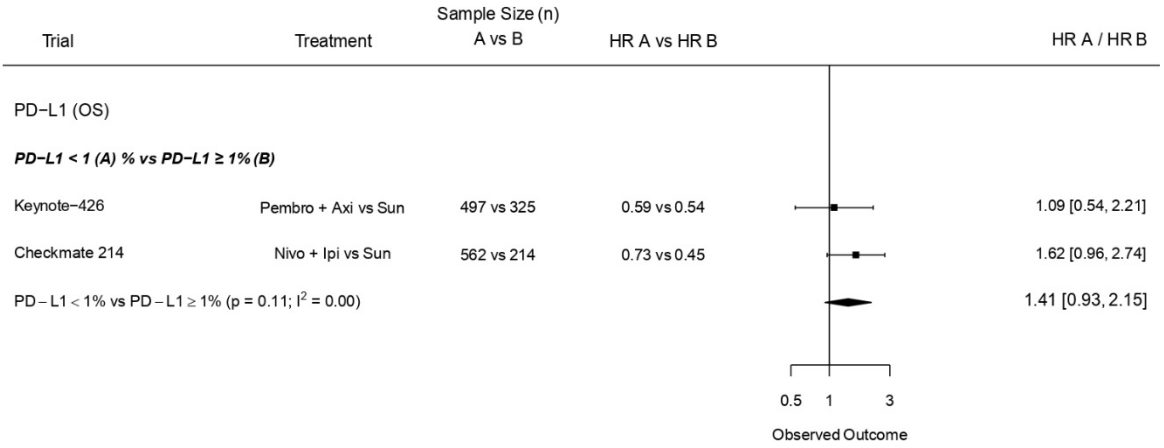

eFigure 15. Fixed-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for MSKCC risk score ($HR_{Favourable}$ to $HR_{Poor}$=1.53; 95% CI:1.00-2.34; $I^2$=0%; *p*=0.05. $HR_{Favourable}$ to $HR_{Intermediate}$=0.96; 95% CI:0.70-1.30; $I^2$=0%; *p*=0.77).

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## PD-L1 (OS)



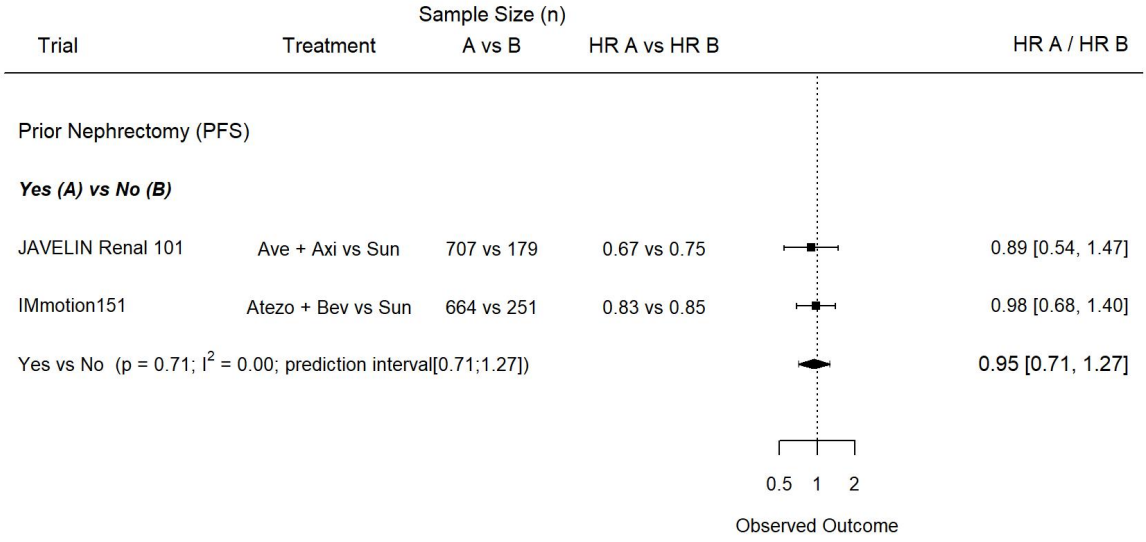

eFigure 16. Random-effects meta-analysis on the ratio of subgroup-specific, overall-survival hazard ratios for PD-L1 ($HR_{PD\text{-}L1<1\%}$ to $HR_{PD\text{-}L1\geq1\%}$=1.41; 95% CI:0.93-2.15; 95% PI:0.93-2.15; $I^2$=0%; *p*=0.11).



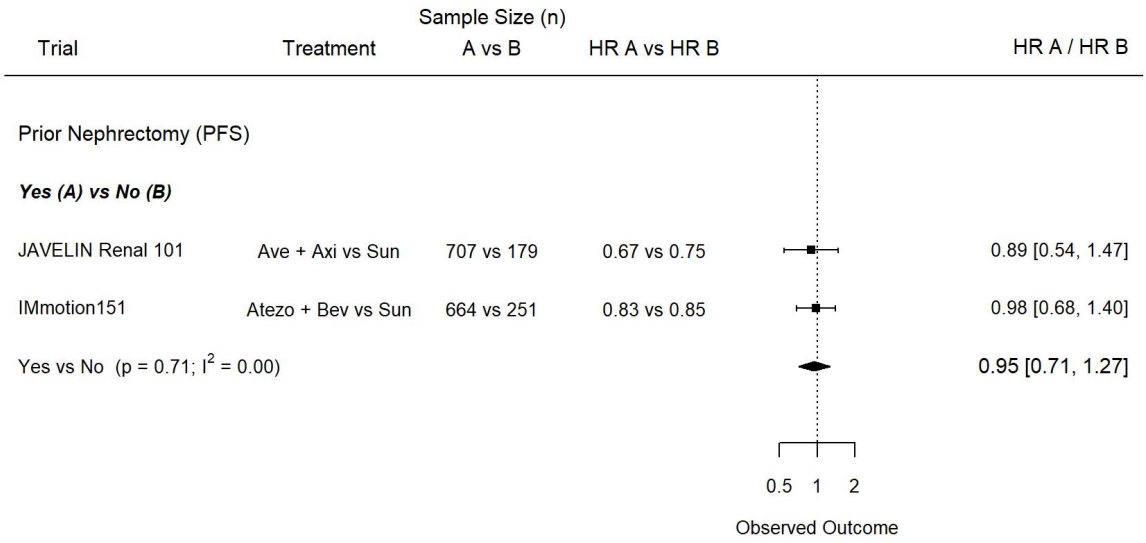

eFigure 17. Fixed-effects meta-analysis on the ratio of subgroup-specific, overall-survival hazard ratios for PD-L1 ($HR_{PD\text{-}L1<1\%}$ to $HR_{PD\text{-}L1\geq1\%}$=1.41; 95% CI:0.93-2.15; $I^2$=0%; *p*=0.11).

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## Prior Nephrectomy (PFS)



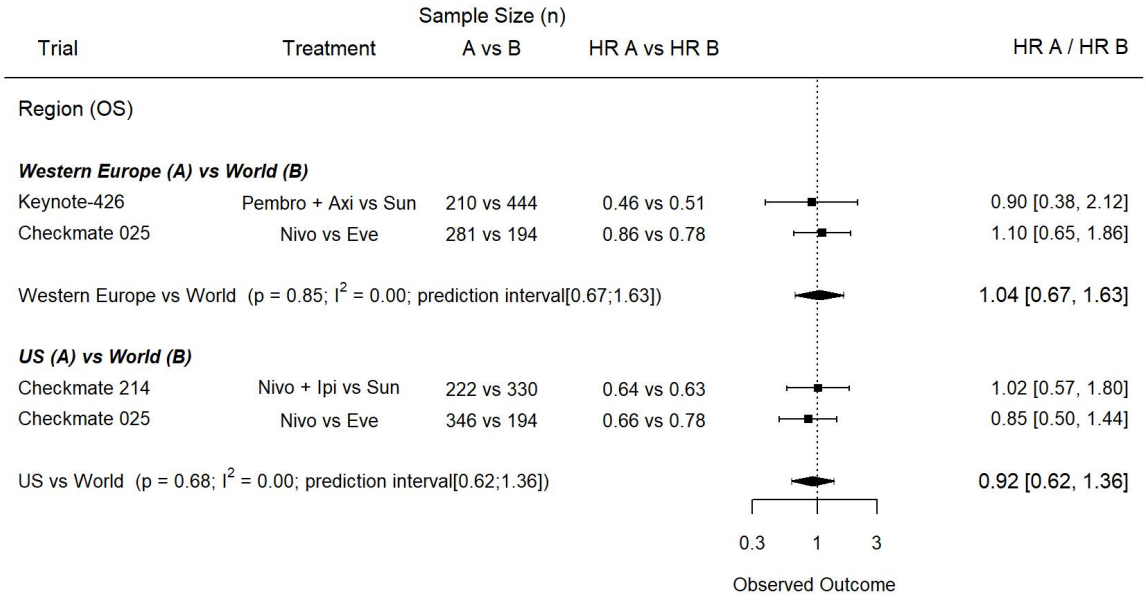

eFigure 18. Random-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for prior nephrectomy ($HR_{Yes}$ to $HR_{No}$=0.95; 95% CI:0.71-1.27; 95% PI:0.71-1.27; $I^2$=0%; *p*=0.71).



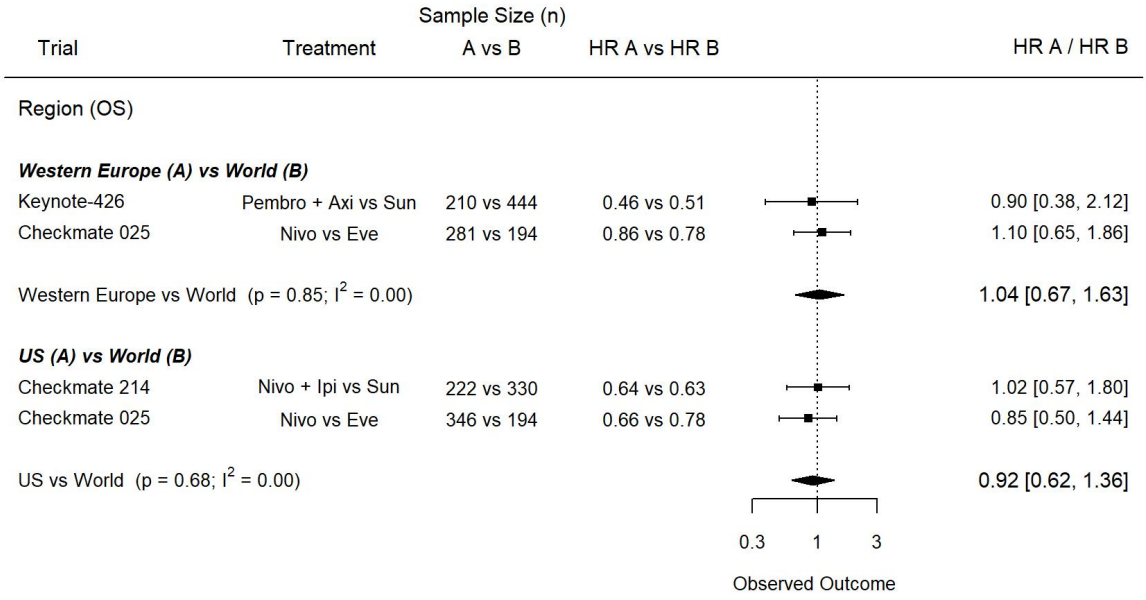

eFigure 19. Fixed-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for prior nephrectomy ($HR_{Yes}$ to $HR_{No}$=0.95; 95% CI:0.71-1.27; $I^2$=0%; *p*=0.71).

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## Region (OS)



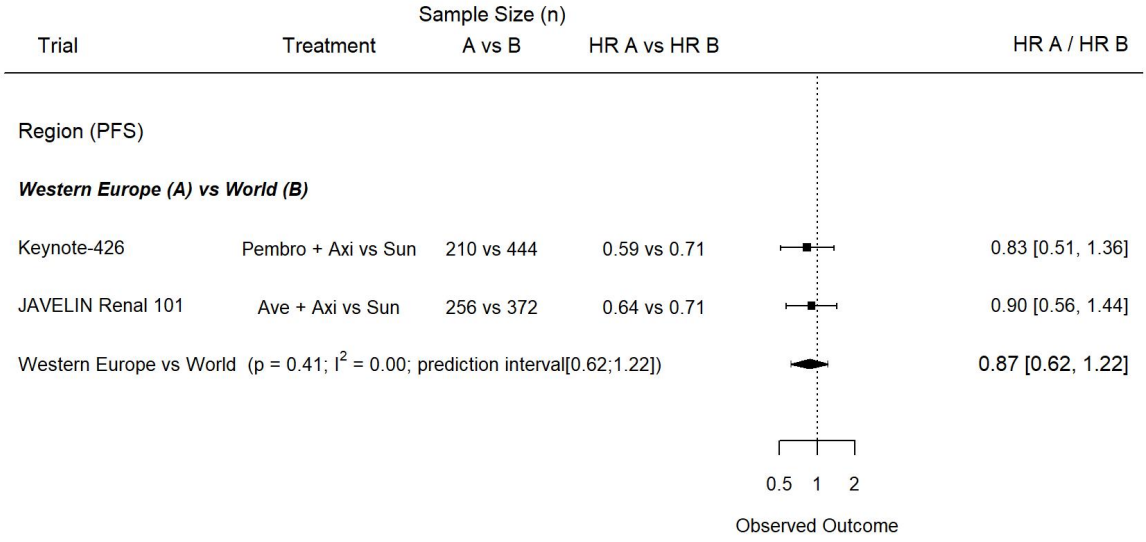

eFigure 20. Random-effects meta-analysis on the ratio of subgroup-specific, overall-survival hazard ratios for region ($HR_{Western\ Europe}$ to $HR_{World}$=1.04; 95% CI:0.67-1.63; 95% PI:0.67-1.63; $I^2$=0%; *p*=0.85; $HR_{US}$ to $HR_{World}$=0.92; 95% CI:0.62-1.36; 95% PI:0.62-1.36; $I^2$=0%; *p*=0.68).



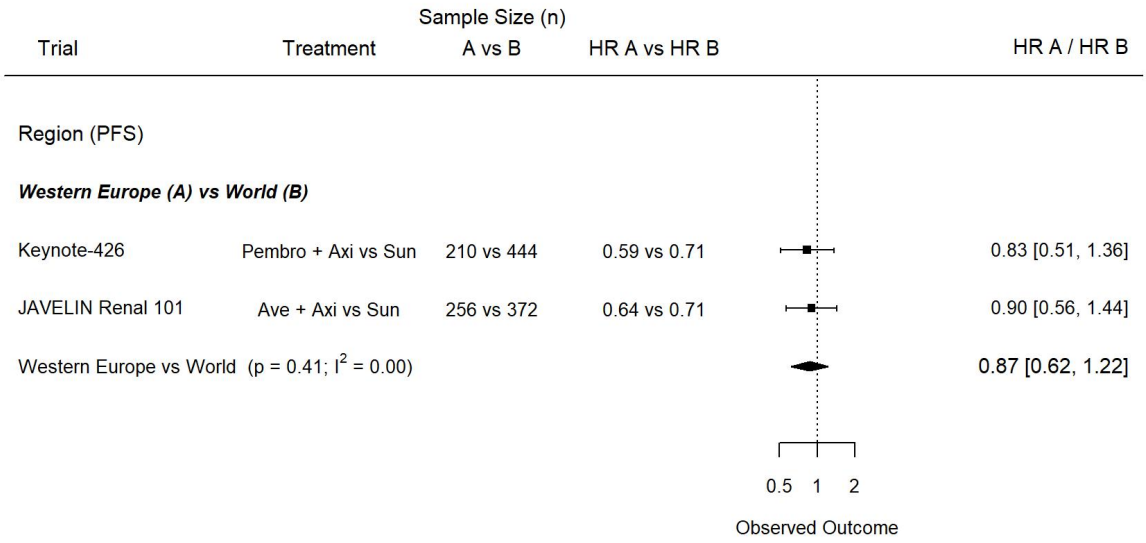

eFigure 21. Fixed-effects meta-analysis on the ratio of subgroup-specific, overall-survival hazard ratios for region ($HR_{Western\ Europe}$ to $HR_{World}$=1.04; 95% CI:0.67-1.63; $I^2$=0%; *p*=0.85; $HR_{US}$ to $HR_{World}$=0.92; 95% CI:0.62-1.36; $I^2$=0%; *p*=0.68).

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## Region (PFS)



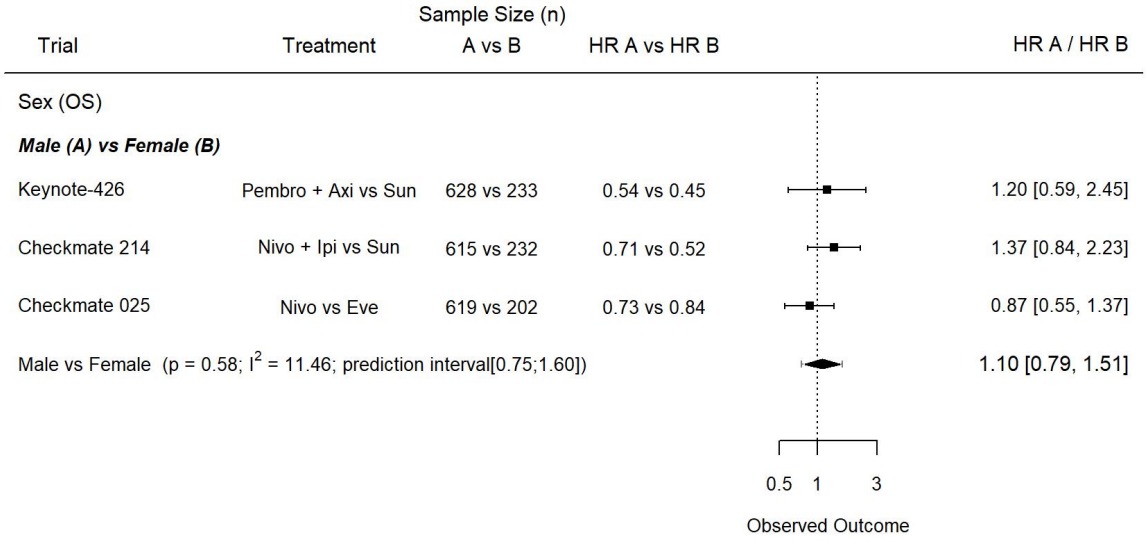

eFigure 22. Random-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for region ($HR_{Western\ Europe}$ to $HR_{World}$=0.87; 95% CI:0.62-1.22; 95% PI: 0.62-1.22; $I^2$=0%; *p*=0.41).



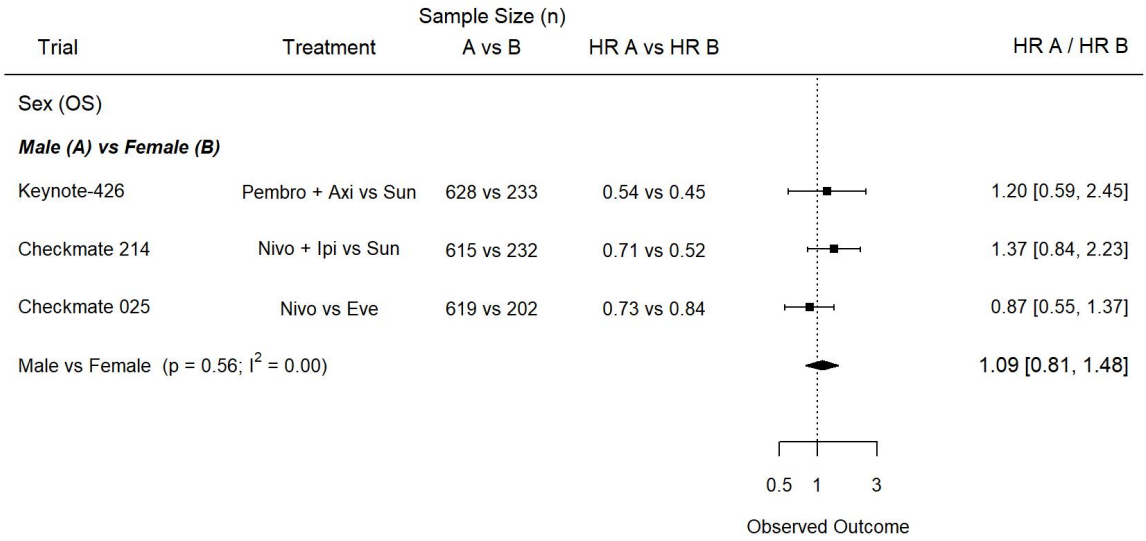

eFigure 23. Fixed-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for region ($HR_{Western\ Europe}$ to $HR_{World}$=0.87; 95% CI:0.62-1.22; $I^2$=0%; *p*=0.41).

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## Sex (OS)



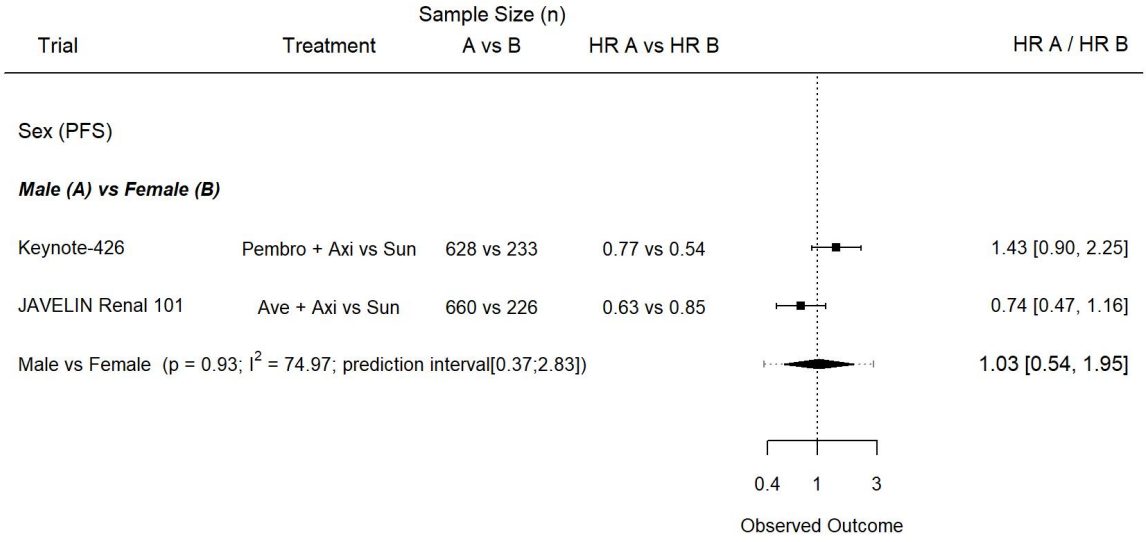

eFigure 24. Random-effects meta-analysis on the ratio of subgroup-specific, overall-survival hazard ratios for sex ($HR_{Male}$ to $HR_{Female}$=1.10; 95% CI:0.79-1.51; 95% PI:0.75-1.60; $I^2$=11.46%; *p*=0.58).



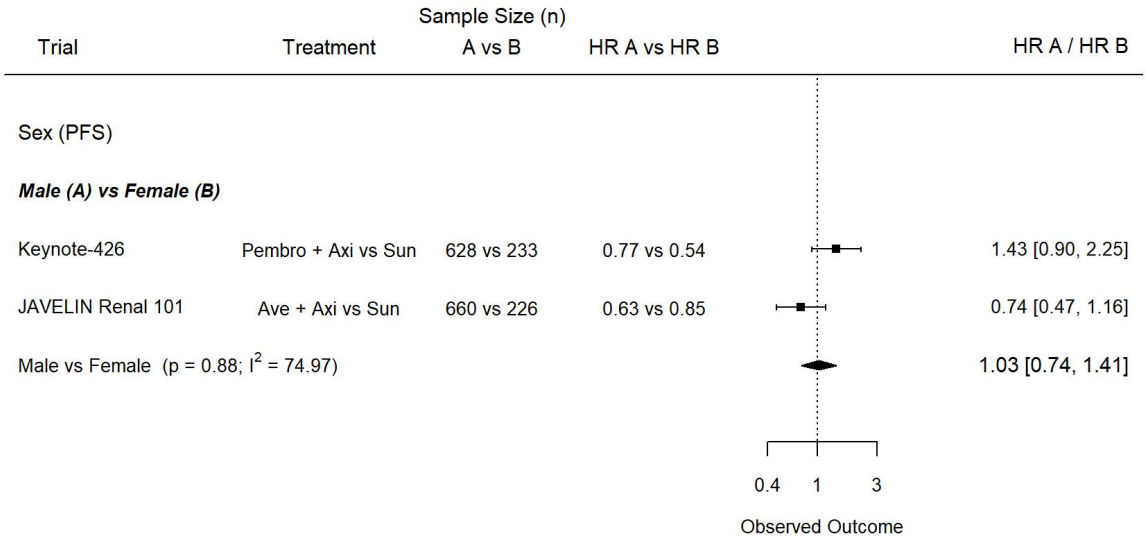

eFigure 25. Fixed-effects meta-analysis on the ratio of subgroup-specific, overall-survival hazard ratios for sex ($HR_{Male}$ to $HR_{Female}$=1.09; 95% CI:0.81-1.48; $I^2$=0%; *p*=0.56).

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## Sex (PFS)



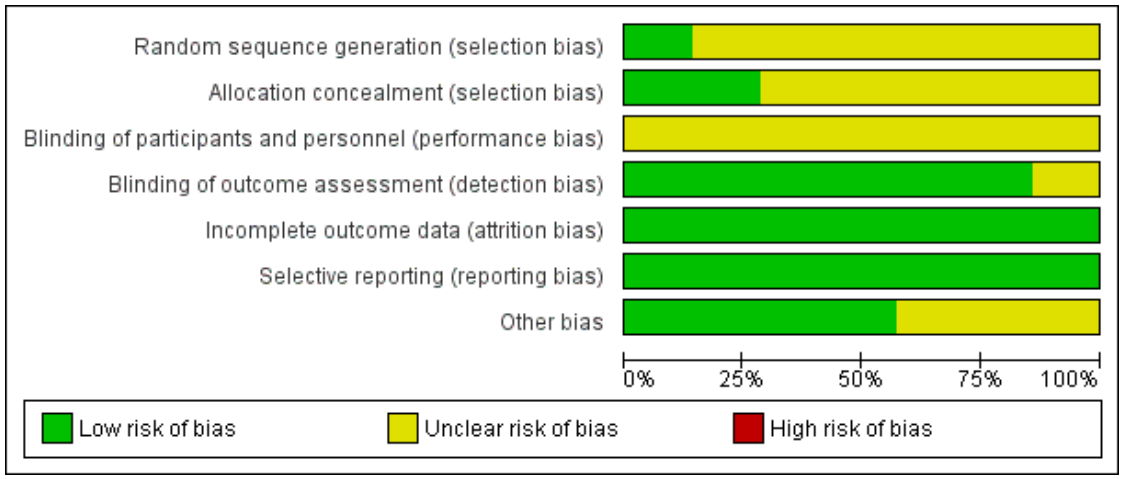

eFigure 26. Random-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for sex ($HR_{Male}$ to $HR_{Female}$=1.03; 95% CI:0.54-1.95; 95% PI:0.37-2.83; $I^2$=74.97%; *p*=0.93).




eFigure 27. Fixed-effects meta-analysis on the ratio of subgroup-specific, progression-free survival hazard ratios for sex ($HR_{Male}$ to $HR_{Female}$=1.03; 95% CI:0.74-1.41; $I^2$=74.97%; *p*=0.88).

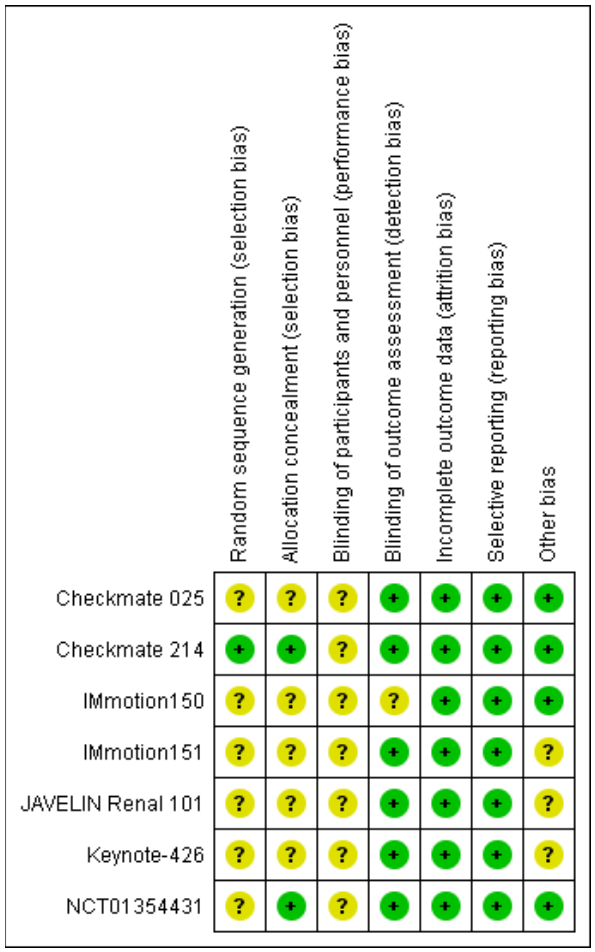
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## Risk of bias (RoB2) assessments




eFigure 28: Risk of bias assessed by review author's judgement of each included trial.

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eFigure 29: Risk of bias summary: review author's judgement of each included trial.

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